



University students' perceptions of the impact of artificial intelligence in the business sector on their educational and professional development

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Received: 16 October 2024 / Accepted: 14 July 2025 / Published online: 29 August 2025
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Abstract

Understanding students' perceptions of Artificial Intelligence (AI) is crucial for integrating AI into education and aligning it with professional expectations. This study investigates how university students' perceptions of AI influence their educational and professional development, a previously unexplored relationship, emphasizing the demand for AI training through Partial Least Squares Structural Equation Modeling (PLS-SEM) analysis on 725 students. The present research offers novel insights into the mediating effects of professional development and the nonlinear role of fear in shaping AI training demand. Findings indicate that students view AI implementation in companies as enhancing business performance through automation and optimized decision-making, thus improving learning and professional development opportunities. High-performing companies are seen as offering valuable internships and motivating students to enhance their skills, thereby increasing the demand for AI training. Students with prior AI knowledge feel more confident and require less additional training. Conversely, perceived risks associated with AI may incentivize academic engagement under certain conditions, but excessive fear may cause mental blocks, reducing participation in educational programs and interest in AI training. This study provides a comprehensive view of how students' perceptions of AI affect their academic and professional paths. These findings offer concrete guidance for curriculum designers and educators aiming to foster AI literacy and pedagogical strategies aligned with technological transformation in business and education. It highlights the importance of educational strategies that incorporate realistic applications and limitations of AI, preparing students for the safe and effective use of technology in their careers. These findings emphasize the need to adapt curricula to the realities of AI in the future workplace.

Keywords Students perceptions on AI · Educational development · Higher education institutions · Professional development · Artificial intelligence ·

Companies · Partial least squares structural equation modelling · PLS-SEM · Nonlinear mediation · Technology adoption · AI education · Professional trajectories

1 Introduction

Artificial Intelligence (AI) is transforming multiple sectors, including education and business, by enabling automation, data-driven decision-making, and personalized learning experiences (Hirsch-Kreinsen, 2024; Keles & Aydin, 2021; Maghsudi et al., 2021; Sagodi et al., 2024). In education, AI has the potential to enhance teaching methods, optimize administrative processes, and adapt learning paths to individual student needs (Chan & Hu, 2023; Popenici & Kerr, 2017). Similarly, in business environments, AI-driven technologies improve operational efficiency, increase productivity, and support strategic decision-making (Abousaber & Abdalla, 2023; Glauner, 2020). Given its growing relevance, AI's integration in education and the job market depends on students' familiarity with these technologies and their preparedness to engage with them.

However, despite AI's increasing presence in both academic and professional settings, there remains a significant gap in understanding how students perceive its role in their education and career development (Almaraz-López et al., 2023; Alzahrani, 2023; Ravi Kumar & Raman, 2022). While previous research has extensively documented AI's benefits for corporate efficiency and automation (Nazaretsky et al., 2022), little attention has been given to how students interpret these advancements and how their perceptions shape their academic choices and professional aspirations. Investigating these perceptions is crucial, as they may influence students' willingness to acquire AI-related skills and adapt to an evolving job market increasingly shaped by AI-driven processes.

A key aspect influencing students' engagement with AI is their perceived impact of AI on business performance. The automation of processes and optimization of decision-making are widely recognized as factors that enhance business success (Thaduri, 2020), and high-performing companies often provide better internship programs and learning opportunities (Gault et al., 2010; León-Gómez et al., 2023). This environment fosters a greater demand for AI-related training, as students recognize the need to develop competencies aligned with evolving industry expectations (Cavallo et al., 2023).

Another critical factor shaping students' approach to AI is their level of prior knowledge and fear of AI. While familiarity with AI can reduce uncertainty and increase confidence in its application (Laupichler et al., 2022), fear of AI may have a dual effect: it can either encourage students to seek more training to better understand and control its impact (Grasse et al., 2023; Pisica et al., 2023) or, conversely, create psychological barriers that deter engagement with the technology (Zucker & Gotian, 2022). This study aims to explore these contrasting influences to provide a more nuanced understanding of how AI perceptions shape students' learning choices.

This research investigates how students' perceptions of AI implementation, particularly in business contexts, inform their educational motivations and expectations,

with implications for curriculum design, AI literacy, and pedagogical innovation in higher education. As future professionals and lifelong learners, students' attitudes toward AI reveal key signals about the adequacy of current academic programs to prepare them for emerging technological environments. By understanding how AI in the workplace shapes their educational choices, this study contributes to the redesign of academic curricula that integrate AI-related content, promote digital competence, and reduce emotional barriers such as fear.

To address these research questions, structural equation modelling using the partial least squares structural equation modelling (PLS-SEM) method was applied to a sample of 725 university students.

This study contributes to the field of educational technology by highlighting how perceptions of AI in the business sector shape students' academic engagement, motivation, and demand for AI-related education. The findings reveal that professional development acts as a key mediator between AI implementation and the desire for further academic training, underlining the role of educational institutions in fostering future-ready skills. These insights offer actionable implications for educators and curriculum designers, who must incorporate AI literacy, practical applications, and emotional preparedness into learning environments. Aligning higher education with the realities of AI-driven workplaces requires rethinking not only what students learn, but how they are taught to navigate and co-evolve with intelligent technologies.

The introductory section is followed by Sect. 2, in which the hypotheses are developed theoretically. Section 3 presents the methodological aspects, and Sect. 4 presents the results. Finally, Sects. 5 and 6 discuss the results and the main conclusions.

2 Theoretical framework

Although various theoretical perspectives may help explain students' responses to artificial intelligence, this study is anchored primarily in two core frameworks to ensure conceptual clarity and analytical consistency. First, Expectancy Value Theory (Eccles & Wigfield, 2002) provides a robust explanation of students' motivation to pursue AI related academic training. Second, the emotional construct of fear is examined using a multidimensional lens, distinguishing between constructive and obstructive responses to technological change. Other perspectives such as Social Cognitive Theory and Self-Determination Theory are acknowledged in the literature but are not used to structure the model or guide hypotheses.

In educational contexts, AI literacy, understood as the ability to critically understand, interact with, and design AI systems, is becoming a fundamental competence for students across disciplines. As AI becomes more embedded in the business world, educational institutions must ensure that curricula evolve accordingly, integrating AI-related content not only in technical degrees, but also in business, humanities, and social sciences (Long & Magerko, 2020). Understanding how students perceive AI implementation in the workplace can guide curriculum developers and educators to anticipate emerging training demands, design more relevant learning outcomes, and reduce emotional barriers such as fear or resistance (Sperling et al., 2024; Zawacki-Richter et al., 2019).

Recent advances in educational technology have emphasized the value of gamified intelligent tutoring systems (GITS) as a means of enhancing student engagement, motivation, and learning outcomes in AI-driven environments. According to a systematic review by Lampropoulos (2025), GITS effectively combine adaptive feedback, game-based mechanics, and personalized pathways to support deeper learning and higher retention across diverse disciplines. Furthermore, a meta-analysis by Batsaikhan & Correia (2024) confirmed that the integration of gamification elements in intelligent tutoring systems significantly improves learners' intrinsic motivation and academic performance, particularly when aligned with individual learning needs. These findings underscore the relevance of incorporating motivational strategies, such as gamification, into AI based educational tools reinforcing the potential for curricular innovation and pedagogical redesign in higher education.

The impact of AI on business performance has been a topic of growing interest in academic research and professional practice (Desta & Amantie, 2024). AI has become an essential tool for many businesses due to its capabilities to automate processes, analyze large volumes of data, and improve decision-making (Thaduri, 2020).

As future employees and business leaders, students are crucial in adopting and developing new technologies within organisations (Biclesanu et al., 2023; León-Gómez & Forero, 2022). Their perception of AI can significantly influence their willingness to work with the technology and their ability to take advantage of its benefits (Chan & Hu, 2023; Idroes et al., 2023). Several studies have shown that individuals' perception of technology influences its acceptance and use (Harst et al., 2019; Williams et al., 2015). Therefore, understanding how students perceive AI and its impact on business performance is essential to foresee how this technology will be integrated into the future work environment (Giunta, 2021). The perceived enhancement of firm performance by AI aligns with theories of technological diffusion, where early exposure reinforces positive attitudes toward adoption (Rogers, 2003). We therefore formulate the following research hypothesis:

Hypothesis 1: Students perceive AI adoption in firms as a driver for improved organizational performance.

Research indicates that, in general, students perceive AI positively and recognise its potential impact on their professional development (Nazaretsky et al., 2022). Students with excellent AI knowledge tend to have more positive attitudes towards its integration into education and future careers (Marrone et al., 2022). Many students believe AI will positively influence society and are willing to further their AI education despite limited current knowledge due to lack of training (Almaraz-López et al., 2023; Gherheş & Obrad, 2018). However, there are some concerns about the rapid development of AI and its effects on humanity (Jeffrey, 2020). Students view the relationship of AI to creativity through social, affective, technological and learning factors, and most believe that human creativity surpasses AI capabilities (Marrone et al., 2022). To address these perceptions, higher education institutions should adapt their curricula to include realistic AI use cases and limitations, preparing students for the safe and responsible use of AI in their future careers (Almaraz-López et al., 2023).

The literature presents a significant gap regarding the perception of students and future professionals in digital transformation and its impact on their professional development. While several studies have explored the benefits of artificial intelligence from a corporate perspective, highlighting increases in efficiency, productivity, and decision-making capabilities (Abousaber & Abdalla, 2023; Glauner, 2020), the student's perspective needs to be more researched. Accordingly, we develop the following research hypothesis:

Hypothesis 2: Students perceive that implementing artificial intelligence in companies positively impacts their professional development.

Business performance understood as a company's ability to achieve its financial and operational objectives (González-Benito & Suárez-González, 2010), directly influences the development of interns, with high-performing companies offering the best learning and networking opportunities (Gault et al., 2010; Hergert, 2009). Factors such as company reputation, quality of educational opportunities and organisational culture play an essential role in students' perception and learning experience (Ahmed et al., 2019; Bell, 2013; Tripathee, 2018).

Several interconnected theories explain how the perception of business performance influences students' professional development. Social Cognitive Theory posits that observational learning from high-performing AI-driven businesses fosters learning behaviors among students, enhancing professional readiness, while Expectancy-Value Theory underscores the motivational pull of perceived professional gains from AI integration (Schunk, 2020; Wang et al., 2023a, b). Indeed, Social Cognitive and Social Learning Theory highlight that imitating successful companies shapes behaviours and motivates them (Manz & Sims, 1981; Schunk, 2012). According to Expectancy-Value Theory and Experiential Learning, high-performance environments improve students' skills and satisfaction and enhance their motivation and preparation for future career challenges (May, 2017; Montalbo, 2022; Renninger & Hidi, 2019; Wang, 2022).

Students' interaction with high-performing companies enhances their practical skills, provides tangible resources, and profoundly influences their psychological and motivational development (D'Abate, 2010; Gallardo et al., 2009; Nada & Ubaidillah, 2024). These experiences internalise successful professional behaviours and strengthen their academic and professional development commitment (Lima et al., 2014). Therefore, further research on how these environments impact the holistic development of students is essential to optimise educational and business practices. Thus, we formulate the following research hypothesis:

Hypothesis 3: Students perceive that the performance of companies favourably affects their professional development.

As students recognise the importance of developing professional skills and competencies, their interest in receiving more extensive and specialised academic training also increases (Chinoy et al., 2022; Kurowska-Pysz, 2014). *Perception of professional development* refers to how students value and understand career opportunities

and advancement due to their academic training (Canet-Vélez et al., 2021; Ganapati & Ritchie, 2021). *Demand for academic training* is defined as the need or interest in acquiring more knowledge, skills and educational credentials to improve their position in the labour market (Stanić et al., 2021; Tomlinson, 2008). To explain this relationship, we consider the theory of Becker (1975) human capital, which argues that investment in education improves skills and career opportunities, and the theory of Deci and Ryan (1985) self-determination, which suggests that intrinsic motivation, based on competence and autonomy, drives the pursuit of lifelong learning and mastery of new areas. This reinforces the importance of rethinking curriculum design in higher education to align with students' evolving professional expectations in AI driven contexts.

This perception is motivated by the need to adapt to a constantly evolving labour market, where lifelong learning is essential to maintain and improve professional competitiveness (Fischer, 2000). Therefore, students who value their professional development are expected to seek further academic training opportunities (Cavallo et al., 2023). Positive perceptions of career development are motivated by both extrinsic reasons, such as better salaries, and intrinsic reasons, such as personal and professional growth (Hirschi, 2010; Lei, 2010). However, most research focuses on specific contexts, such as certain professions or levels of education, leaving a gap in understanding these dynamics in a broader and more diverse student population (Bostock & Baume, 2016; Fahnert, 2015; Nychkalo et al., 2022). Thus, we formulated the following research hypothesis:

Hypothesis 4: Students perceive that their professional development fosters a greater demand for IA academic training.

Their prior AI knowledge influences students' perception of the need for additional academic training (Almaraz-López et al., 2023). This prior knowledge refers to the degree of familiarity and understanding of AI concepts, techniques and applications prior to formal academic training (Jyothyachandra & Sulaimann, 2022; Lindner & Berges, 2020). The perceived need for further academic training is the subjective assessment of the importance of continuing their formal AI education (Leao et al., 2018; Maity, 2019).

According to constructivist learning theory, prior knowledge facilitates understanding new concepts, making students with a solid foundation in AI feel competent without needing additional training (Morphew, 2002). Wood and Lynch (2002) suggests that students with advanced knowledge in a specific area value additional academic training less. This relationship is based on prior knowledge facilitating autonomous learning and self-efficacy (Beatson et al., 2020). However, in the field of AI, due to rapid technological advancement (Jha et al., 2022), this may not be the case. Students who have mastered fundamental AI concepts may prefer to keep up to date and acquire new knowledge on a self-taught or hands-on basis rather than resorting to additional formal courses (Amershi et al., 2005; Aunimo & Huttunen, 2020). Those with a solid foundation in AI may feel competent to face future challenges, thus decreasing their perception of the need for additional training (Straub et al., 2017).

The scientific literature published so far has yet to conclusively specify whether prior knowledge of AI affects the demand for further academic training in this field (Laupichler et al., 2022). To date, research has focused on the role of prior knowledge in academic success within specific courses rather than investigating how this knowledge influences the perceived need for additional AI training (Hailikari et al., 2008; Hailikari & Nevgi, 2010; Seery, 2009). Therefore, there is a significant opportunity to examine whether or not students perceive that their prior AI knowledge reduces the demand for additional academic training, thus filling a crucial gap in the existing literature.

Hypothesis 5: Students perceive that their prior knowledge of Artificial Intelligence reduces the demand for IA academic training.

AI research is an ongoing concern within academic education. (Oluwadiya et al., 2023). Fear of AI refers to the apprehension, anxiety, or concern individuals or groups may have about the potential consequences and implications of AI advancements (Johnson & Verdicchio, 2017). Fear of change theory and social learning theory can explain this relationship (Bagozzi et al., 1992). Fear of change theory posits that uncertainty and the perceived threat of new technologies motivate individuals to seek ways to adapt and mitigate them (Alhumaid, 2021). On the other hand, social learning theory argues that people adopt behaviours and seek knowledge by observing others in their environment (Tscherning, 2012). These theories suggest that fear of AI motivates people to seek AI education to cope with uncertainty and feel safer (Raza et al., 2021).

The relationship between fear of AI and increased demand for academic education in AI is explained by the need to reduce uncertainty and increase professional competence (Pisica et al., 2023). Fear of AI generates a perception of vulnerability to possible occupational and social disadvantages (Zawacki-Richter et al., 2019). By acquiring AI knowledge and skills, people can better control their professional future and reduce the risk of becoming obsolete (Alekseeva et al., 2019). In addition, AI education provides a sense of security and confidence, mitigating the fear of the unknown and the possible negative impacts of artificial intelligence (Grasse et al., 2023).

Fear of AI, grounded in Fear of Change Theory, can paradoxically drive training motivation by reducing uncertainty (Alhumaid, 2021). By integrating Fear of Change Theory into AI education research, this study offers a novel lens to understand emotional barriers in technology adoption.

However, when perceived fear is too high, it can cause mental blocks that prevent people from engaging in activities they consider threatening (Zucker & Gotian, 2022). According to anxiety processing theory, fear can block people's ability to process information rationally, resulting in a decrease in decision-making and taking on new challenges (Beck & Clark, 1997). Consequently, this theory suggests that an elevated level of fear may inhibit interest and participation in AI-related educational programmes (Beck & Clark, 1997; Fofana et al., 2020).

These mental blocks can reduce self-esteem and creativity, limiting individuals' ability to make informed decisions and take on new challenges (Hu et al., 2021;

Zucker & Gotian, 2022). In AI training, fear can be so intense that it not only deters people from enrolling in courses but can also reduce demand for training as fear reaches critical levels, completely blocking the learning initiative (Alexandre et al., 2021; Hsu et al., 2023; Lemay et al., 2020).

To date, research has mainly focused on the impact of fear of AI on the demand for AI training (Alexandre et al., 2021; Hsu et al., 2023; Zucker & Gotian, 2022). However, few studies have examined how extremely high levels of fear can decrease interest and participation in AI training. This lack of specific studies leads us to pose the following research hypotheses:

Hypothesis 6: Students perceive fear of Artificial Intelligence as increasing demand for academic training.

Hypothesis 7: The effect of fear of AI on the demand for AI training decreases when fear levels reach high levels.

The rapid adoption of AI in the business world is transforming internal operations and the perspectives and needs for academic training among students (Cubric, 2020). This is because introducing AI in business first creates a need for new technological competencies and skills, which motivates students to seek academic training that prepares them for these new challenges (Rubab, 2023). Second, students recognise that AI skills give them a competitive advantage in the labour market, which incentivises their interest in specialised educational programmes (Chen et al., 2022). Third, students' professional development is enhanced by the job and growth opportunities that arise in companies that adopt AI, increasing the demand for academic training (Kamaruddin et al., 2023; Okello, 2023). However, it has yet to be elucidated how students' perceptions of their professional development may mediate the relationship between AI implementation in companies and the demand for academic training. Therefore, we propose the following research hypothesis:

Hypothesis 8: Students perceive that implementing artificial intelligence in companies indirectly enhances the demand for academic training by boosting their professional development.

Research on the impact of high-performing companies on students' professional development shows that these successful organisations are better positioned to offer resources and educational opportunities (Johnson & Thomas, 2010). These companies have robust professional development programmes, leading to significant practical experiences for students, enhancing their technical and soft skills, and providing them with a clear understanding of the expectations and demands of the labour market (Bayraktaroğlu & Cickusic, 2013; Shaha et al., 2015). As a result, students who participate in these programmes are better prepared to face professional challenges and advance in their careers (Bogoviz et al., 2019).

However, the literature needs to pay more attention to how the adoption of AI in companies may indirectly influence students' perceptions of the need for academic training (Popenici & Kerr, 2017; Zawacki-Richter et al., 2019). The implementation of AI improves the operational efficiency of companies by automating processes,

reducing human error, optimising decision-making, and personalising products and services (Kishan Kumar, 2024; Zyryanova et al., 2024). These improvements lead to superior business performance, increased productivity, higher revenues, and sustained competitive advantage (Kaleka & Morgan, 2017; Singh, 2012). Observing these positive outcomes, students conclude that companies adopting AI seek employees with advanced skills and specific technology-related knowledge (Malik et al., 2021; Meshram, 2023).

This perception generates among students the need to increase their academic training in relevant areas such as data science, artificial intelligence, and technology management to access better job opportunities and be competitive in a technology-driven market (Apraksin et al., 2018; Johnson et al., 2021). Nonetheless, research has focused on AI's direct impact on companies' operational efficiency and has yet to explore how these technological advances affect students' perceptions and educational decisions (Alzahrani, 2023; Chen et al., 2020).

Despite this, some studies have found that students' perception of AI implementation in companies impacts the demand for academic training (Bogoviz et al., 2019; Marcinkowski et al., 2020). AI enhances business performance by optimising processes and improving efficiency, creating a competitive environment students see as a professional opportunity (Pisica et al., 2023; Wisniewski, 2020). Therefore, students seek to acquire the necessary skills for these technological environments, perceiving that AI-driven business success increases their interest in training in technological areas, thus ensuring their competitiveness in the labour market (Jha et al., 2022; Lee et al., 2023; Matthews & Volpe, 2023; Nam & Ha, 2023).

The generalisation of research on the implementation of AI in companies and its impact on the demand for academic training is problematic due to the need for more specific studies (Bogoviz et al., 2019). Research has mainly focused on operational efficiency and productivity, neglecting its influence on professional and educational development (Nurlia et al., 2023; Zakaria et al., 2019). Exploring this relationship is a valuable opportunity to better understand the dynamics between advanced technology and education, improving business performance and professional development (Tanveer et al., 2020). For this reason, we formulate the following research hypotheses:

Hypothesis 9: Students perceive that implementing artificial intelligence in companies improves organizational performance and, in turn, contributes to their professional development.

Hypothesis 10: Students perceive that companies' adoption of artificial intelligence indirectly leads to an increased demand for IA academic training through improvements in these companies' performance.

Hypothesis 11: Students perceive that implementing artificial intelligence in companies indirectly stimulates an increase in the demand for IA academic training by elevating the performance of the companies and consequently enriching their professional development.

In short, students' perceptions of AI in business do not simply reflect labour market trends, but act as a mirror of what higher education is (or is not) providing in terms

of preparation. These perceptions demand a pedagogical response: the integration of AI literacy, emotional readiness and interdisciplinary competence into academic programmes. For this reason, we have established the following model, which is represented in Fig. 1.

3 Methods

3.1 Data collection

To conduct this research, a structured questionnaire was designed and distributed via institutional email to undergraduate students at the University of Almería, the University of Málaga, and the University of Murcia during the first quarter of 2024. The selection of participants was based on convenience sampling, targeting students enrolled in business, engineering, and education programs related to digital transformation, quantitative methods, or innovation. This approach allowed for efficient access to a relevant population segment, though it carries limitations regarding representativeness. Specifically, convenience sampling may overrepresent students who are more digitally engaged or familiar with artificial intelligence (AI), potentially underrepresenting other perspectives. This could introduce bias and limit the generalizability of the findings to the broader university student population (Etikan, 2016). Despite these limitations, university students were considered a strategically relevant group, as they are future professionals and a key target of AI-related educational innovation (Bornstein et al., 2013). The final sample consisted of 725 valid responses, distributed across the three universities. As shown in Table 1, the sample presents a balanced gender distribution and includes students from diverse academic backgrounds. As can be seen, the gender distribution is quite balanced.

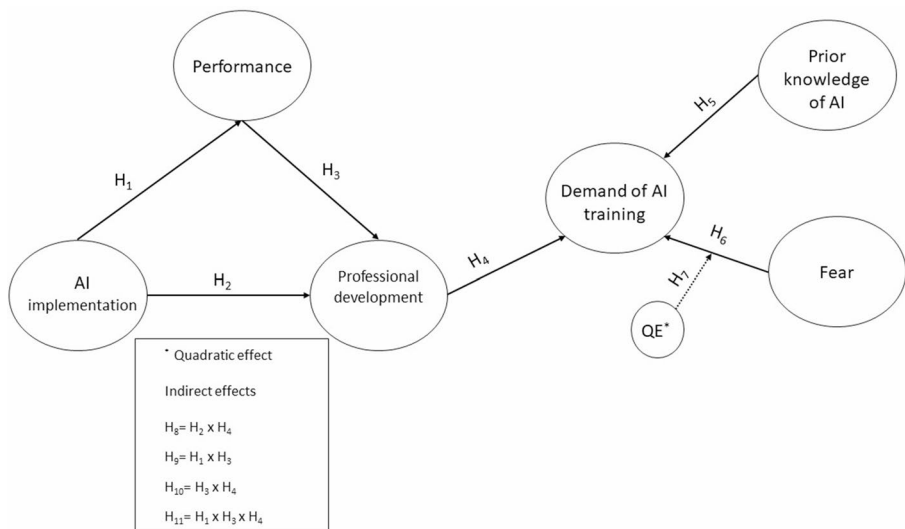


Fig. 1 Proposed model. Source: Own elaboration

Table 1 Sample distribution

University	Man	Woman	Total
University of Murcia	173	149	322
University of Almería	26	43	69
University of Málaga	155	179	334
	354	371	725

Source: Own elaboration

The survey underwent iterative pretesting, with cognitive debriefing conducted to ensure interpretability across demographic subgroups. Specifically, to ensure the survey was easily comprehensible, a pretest was conducted with six reliable students prior to commencing the fieldwork. After obtaining the final sample, checks were carried out to address issues related to non-response bias and common method bias. To tackle non-response bias, the sample was split into two groups (with 75% of the initial responses in one group and the remaining in the other), and an ANOVA test showed no significant differences between their responses. To eliminate common method bias, a single factor test was performed (Podsakoff et al., 2003), revealing that six factors accounted for 63.75% of the total variance, with the main latent factor explaining 34.15%. Thus, it was concluded that common method bias was not an issue, allowing for the successful completion of the fieldwork (Hair et al., 2019). Furthermore, to avoid non-response bias, the questionnaire was completely anonymous.

Finally, to validate the effects discovered in this study, we used G*Power 3.1.9.7 software (Mayr et al., 2007) to assess the appropriateness of the sample size. Considering three connections to the final dependent variable, and assuming an average effect size of 0.15, an alpha level of 0.05, and a power of 0.95, the results indicated that a minimum of 119 observations is required (Cohen, 2013). This required number is significantly lower than the sample size used in the current research. Furthermore, considering the sample size and the total number of university students across the three institutions involved in the study, the sampling error is calculated at 3.65%, with a 95% confidence level.

3.1.1 Measurement variables

To investigate the hypotheses presented in this study, and based on previous literature, six latent variables were established. Three of these variables have been defined as mode A composites (AI implementation, performance and professional development) due to a presumed definitive relationship between the indicators and the latent variables. Mode A (reflective) was also utilized because of the high correlation between the indicators used to construct each variable. The other three variables consist of a single item each. A 5-point Likert scale was used to measure each item making up the latent variables. The variable ‘Prior Knowledge of AI’ was measured through self-assessment, meaning it captures participants’ perceived knowledge rather than an objective evaluation of their technical understanding. Table 2 displays the composition and explanation of the variables used in this study.

Table 2 Variables used in the study**AI implementation**

Indicate from 1 to 5, where 1 is not at all and 5 is very much, the degree of implementation you believe AI will have in the following business departments (Dubey et al., 2020; Reim et al., 2020):

AI_0.1	Human Resources
AI_0.2	Finance
AI_0.3	Management
AI_0.4	Accounting
AI_0.5	Production
AI_0.6	Sales
AI_0.7	Supply
AI_0.8	Marketing
AI_0.9	Research and Development

Performance

Indicate from 1 to 5, where 1 means strongly disagree and 5 means strongly agree, your level of agreement with the following statements (Chatterjee et al., 2021 ; García-Lopera et al., 2022; Gimeno-Arias et al., 2021; Pendy, 2023):

Perf_0.1	AI will increase the sales of companies
Perf_0.2	AI will increase the profitability of companies
Perf_0.3	AI will increase the return on capital employed
Perf_0.4	AI will improve the quality of products and services provided
Perf_0.5	AI will enable companies to develop new products and services

Professional development

Indicate from 1 to 5, where 1 means strongly disagree and 5 means strongly agree, your level of agreement with the following statements (Allen et al., 2021; Grassini, 2023; Sard et al., 2019):

Pd_0.1	I believe that my professional future will be influenced by AI
Pd_0.2	The implementation of AI represents a professional stimulus for me
Pd_0.3	AI will provide a professional advantage for me

Prior Knowledge of AI

Indicate from 1 to 5, where 1 means strongly disagree and 5 means strongly agree, your level of agreement with the following statement (Chetty, 2023; Wang et al., 2023):

Pk_0.1	I have sufficient knowledge about AI
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Fear

Indicate from 1 to 5, where 1 means strongly disagree and 5 means strongly agree, your level of agreement with the following statement (Moradi & Levy, 2020; Vorobeva et al., 2022; Walusiak-Skorupa et al., 2023):

Table 2 (continued)

F_0.1	The development of AI makes me fearful
Demand of AI training	
<i>Indicate from 1 to 5, where 1 means strongly disagree and 5 means strongly agree, your level of agreement with the following statement (Morandini et al., 2023; Zhao, 2021):</i>	
Dt_0.1	I should train more in AI

Source: Own elaboration

3.1.2 Statistical procedure

In the previous section, we noted that our model includes three type A composites. Based on the recommendations of Cepeda-Carrion et al. (2019) and Hair et al. (2019), PLS-SEM is the most appropriate technique for analyzing the relationships between these variables. PLS-SEM is especially useful when dealing with multiple relationships that include mediating effects (Cepeda et al., 2024), and when studying nonlinear relationships (Ghasemy et al., 2021), both of which apply to our model.

We applied this model for both confirmatory and explanatory purposes (Sharma et al., 2024) using the SmartPLS 4.1.0.6 software (Ringle et al., 2022). Furthermore, as recommended by Streukens and Leroi-Werelds (2016), the bootstrapping procedure was conducted with 10,000 samples to ensure accuracy.

The model's goodness-of-fit indices, such as SRMR and NFI, were thoroughly evaluated to ensure robustness.

4 Results

4.1 Measurement model analysis

To validate the proposed hypotheses, the measurement model will initially be assessed to ensure the items and variables' validity and reliability. This assessment includes examining indicator reliability, internal consistency reliability, convergent validity, and discriminant validity (Ghasemy et al., 2021).

The reliability of the items used is verified through the analysis of their loadings, as presented in Table 3. The majority of the items exceed the minimum established value of 0.708 (Hair et al., 2024), and those that fall short are very close to this threshold, all being above 0.4. This is a threshold beyond which an item may be retained in the model if deemed significant (Hair et al., 2019). Additionally, the reliability of the construct was evaluated by examining Cronbach's alpha, composite reliability, and the Dijkstra-Henseler rho ratio. Since these values are all above the 0.7 threshold, the construct can be considered reliably measured (Dijkstra & Henseler, 2015). Finally, the average variance extracted (AVE) was used to assess convergent validity, with all values exceeding 0.5, thereby confirming the internal consistency of the constructs (Hair et al., 2021).

Table 4 presents the discriminant validity analysis of the constructs. The values above the diagonal (in italics) represent the HTMT ratios, which are used to assess

Table 3 Measurement model analysis

Composite indicators	Mean	SD	Loading	t-student*	α	ρ_A	ρ_C	AVE
AI implementation					0.828	0.839	0.868	0.517
AI_0.1	3.372	1.172	0.626	24.106***				
AI_0.2	4.003	0.948	0.641	21.138***				
AI_0.3	3.229	1.187	0.694	33.619***				
AI_0.4	3.716	1.266	0.698	31.744***				
AI_0.5	3.513	1.263	0.753	39.694***				
AI_0.6	3.531	1.250	0.745	37.617***				
AI_0.7	4.090	1.073	0.592	18.372***				
AI_0.8	4.150	1.057	0.583	17.284***				
Performance					0.899	0.900	0.925	0.713
PA_1	3.763	1.102	0.871	82.167***				
PA_2	3.902	1.084	0.878	77.434***				
PA_3	3.761	1.134	0.855	62.033***				
PA_4	3.552	1.220	0.808	57.060***				
PA_6	4.003	1.139	0.807	45.283***				
Professional development					0.801	0.802	0.884	0.718
Pd_0.1	3.727	1.236	0.775	37.575***				
Pd_0.2	3.341	1.291	0.883	90.543***				
Pd_0.3	3.510	1.328	0.879	85.215***				
Prior Knowledge of AI								
Pk_0.1	2.250	1.141						
Fear								
F_0.1	2.865	1.332						
Demand of AI training								
Dt_0.1	3.868	1.250						

Significance and standard deviations (SD) performed by 10,000 repetitions Bootstrapping procedure. α : Chronbach's alpha; ρ_A : Dijkstra–Henseler's composite reliability; ρ_C : Jöreskog's composite reliability; AVE: Average Variance Extracted; *: All loadings are significant at the 0.001 level

Source: Own elaboration

Table 4 Discriminant validity analysis

		I	II	III	IV	V	VI
I	AI implementation	0.674	<i>0.805</i>	<i>0.766</i>	<i>0.150</i>	<i>0.163</i>	<i>0.545</i>
II	Performance	0.642	0.844	<i>0.796</i>	<i>0.159</i>	<i>0.203</i>	<i>0.613</i>
III	Professional development	0.633	0.676	0.847	<i>0.243</i>	<i>0.152</i>	<i>0.652</i>
IV	Prior knowledge of AI	0.137	0.150	0.218	1.000	<i>0.019</i>	<i>0.118</i>
V	Fear	0.115	0.193	0.067	−0.019	1.000	<i>0.237</i>
VI	Demand of IA training	0.504	0.581	0.583	0.118	0.237	1.000

HTMT ratio over the diagonal (italics). Fornell–Lacker criterion: square root of AVE in diagonal (bold) and construct correlations below the diagonal

Source: Own elaboration

discriminant validity. HTMT values below 0.85 are typically considered acceptable, indicating that constructs are distinct from each other (Hair et al., 2024). The diagonal values in bold show the square root of the Average Variance Extracted (AVE) for each construct. According to the Fornell–Larcker criterion (Fornell & Larcker, 1981),

these values should be higher than the correlations between the constructs (values below the diagonal) to confirm discriminant validity. Therefore, the results support the discriminant validity of the constructs, and confirm the model's constructs are distinct, addressing potential measurement overlap.

To conclude the analysis of the external model, it has been confirmed that for both the estimated and saturated models, the standardized root mean square residual (SRMR) values are below the specified threshold of 0.08. Similarly, the normed fit index (NFI) values exceed 0.9. These results indicate a satisfactory model fit (Hair et al., 2024; L.-T. Hu & Bentler, 1998).

4.2 Structural model analysis

To begin the analysis of the structural model, it was verified that the Variance Inflation Factor (VIF) between the constructs does not exceed the threshold of 3. As shown in Table 5, the VIF values are well below 3, thereby alleviating any concerns about collinearity issues (Hair et al., 2019).

To accept or reject the proposed hypotheses, the analysis of the structural model continues with the evaluation of the signs, magnitudes, and significance of the established relationships. For this purpose, a one-tailed percentile bootstrapping procedure with 10,000 subsamples has been executed. The results are shown in Table 6.

The analysis of the direct effects shows how the implementation of artificial intelligence positively and significantly influences both performance and professional development ($\beta=0.707^{***}$ and $\beta=0.310^{**}$, respectively), leading us to accept hypotheses 1 and 2. Regarding performance, the results show that it positively and significantly influences professional development ($\beta=0.456^{***}$), thus accepting hypothesis 3. In relation to professional development, it is observed that it positively and significantly influences the demand of AI training ($\beta=0.567^{***}$), which allows us to accept hypothesis 4. On the other hand, the effect of prior knowledge on the demand for AI training is not significant ($\beta=0.004^{ns}$), so hypothesis 5 is rejected. The rejection of H5 suggests that even knowledgeable students may pursue formal training due to the rapidly evolving nature of AI technologies, highlighting the need for continuous learning ecosystems. Finally, it can be seen that fear positively and significantly influences the demand of AI training ($\beta=0.205^{***}$). Therefore, hypothesis 6 is accepted.

This study also investigated the indirect effects and the variance accounted for (VAF) (Hair et al., 2011), which measures the proportion of the indirect effect relative to the total effect (Hair et al., 2014). The results show that AI implementation

Table 5 Variance inflation factor analysis

	I	II	III	IV	V	VI
I	AI implementation	1.000	1.999			
II	Performance		1.999			
III	Professional development					1.061
IV	Prior knowledge of AI					1.070
V	Fear					1.023
VI	Demand of IA training					

Source: Own elaboration

Table 6 Structural model analysis

	Path	SD	T-value	CI		f^2	H	Sup-ported
				5%	95%			
Direct effects								
AI imple-mentation ->Performance	0.707	0.022	32.658***	0.671	0.743	1.028	H ₁	YES
AI imple-mentation ->Professional development	0.310	0.039	7.916***	0.246	0.374	0.094	H ₂	YES
Performance ->Professional development	0.456	0.040	11.500***	0.390	0.521	0.207	H ₃	YES
Professional development ->Demand of AI training	0.567	0.032	17.656***	0.513	0.619	0.490	H ₄	YES
Prior knowl-edge of AI ->Demand of IA training	0.004	0.034	0.126 ^{ns}	-0.052	0.060	0.000	H ₅	NO
Fear ->De-mand of AI training	0.205	0.032	6.495***	0.152	0.256	0.063	H ₆	YES
Indirect effects						VAF		
AI imple-mentation ->Professional development ->Demand of AI training	0.176	0.024	7.374***	0.138	0.216	0.490	H8	YES
AI imple-mentation ->Performance ->Professional development	0.323	0.031	10.481***	0.273	0.374	0.510	H9	YES
Performance ->Professional development ->Demand of AI training	0.259	0.029	8.999***	0.212	0.307	1.000	H10	YES
AI imple-mentation ->Performance ->Professional development ->Demand of AI training	0.183	0.022	8.318***	0.148	0.220	0.509	H11	YES
<i>Global indirect effects</i>								
AI imple-mentation ->Professional development	0.323	0.031	10.481***	0.273	0.374	0.510		

Table 6 (continued)

	Path	SD	T-value	CI		f^2	H	Supported
				5%	95%			
AI implementation ->Demand of AI training	0.359	0.026	13.566***	0.316	0.404	1.000		
Performance ->Demand of AI training	0.259	0.029	8.999***	0.212	0.307	1.000		
Quadratic effects								
QE (Fear) ->06_Demand of AI training	-0.056	0.033	1.667*	-0.110	-0.001	0.002	H7	YES
Total effect								
AI implementation ->Professional development	0.633	0.022	26.364***	0.594	0.673			
AI implementation ->Demand of AI training	0.359	0.026	13.566***	0.316	0.404			
Performance ->Demand of AI training	0.259	0.029	8.999***	0.212	0.307			

R^2 adjusted [95% CI in brackets]: Performance: 0.506 [0.457; 0.557]; Professional development: 0.502 [0.454; 0.553]; Demand of AI training: 0.379 [0.322; 0.444]. SD: Standard Deviation; f^2 : size effect index; QE: Quadratic effect; CI: 95% Bias Corrected Confidence Interval; Significance, standard deviations, 95% bias-corrected CIs were performed by 10,000 repetitions Bootstrapping procedure; *: $p < 0.05$; ***: $p < 0.001$

Source: Own elaboration

indirectly influences the demand of AI training. This occurs through the mediation of professional development ($\beta=0.176^{***}$, VAF=49%) and through the sequential mediation of performance and professional development ($\beta=0.183^{***}$, VAF=51%). This allows us to accept both hypothesis 8 and hypothesis 11. Regarding the indirect effect, since there is no prior direct effect, it is considered a case of full mediation. The findings also illustrate the mediating role of performance in the link between implementation of AI and professional development. The indirect effect is 0.323^{***} (VAF=51%), indicating that the proportion mediated is substantial, but both the direct and indirect effects are significant, suggesting partial mediation, thus accepting the hypothesis 9. Similarly, the indirect effect of performance on demand of AI training through innovation is positive and significant (value= 0.259^{***} , VAF=100%), indicating complete mediation as the direct effect does not exist, supporting the hypothesis 10.

The R^2 analysis reveals that the model explains 51% of the variance in performance, 50% in professional development, and 38% in demand of AI training. According to Hair et al. (2019), R^2 values of 0.75, 0.50, and 0.25 are considered substantial, moderate, and weak, respectively. Therefore, these results confirm the model's satisfactory explanatory power.

To conclude the structural model analysis, we evaluated the effect size (f^2) to determine the contribution of each independent variable to the dependent variable. According to Cohen (1988), a minimum value of 0.02 was used to assess the validity of the hypotheses. Hair et al. (2019) further categorize effect sizes: a small effect exists if the value ranges between 0.02 and 0.15, which we observed in the impact of AI implementation on professional development and the influence of fear on the demand for AI training. A medium effect, identified with values between 0.15 and 0.35, was noted in the relationship between performance and professional development. Finally, an effect size greater than 0.35 signifies a large effect, as seen in the impact of AI implementation on performance and the influence of professional development on the demand for AI training.

Figure 2 shows the results obtained regarding the standardized path coefficients and R^2 values.

5 Analysis of the quadratic effect

In addition, in order to test hypothesis 7, the proposed model has been enhanced with a nonlinear effect through a quadratic function between fear and demand of AI training, using SmartPLS (Basco et al., 2021). For this analysis, a negative quadratic interaction with a positive linear effect on the relationship between fear and demand of AI training was utilized. Additionally, to ascertain the significance of the nonlinear effect, the two-stage approach was adapted for model estimation (Sarstedt et al., 2020).

The findings indicate that the linear effect of fear on demand of AI training increased from 0.199 (Table 6) to 0.205, with an f^2 effect size of 0.063, which is

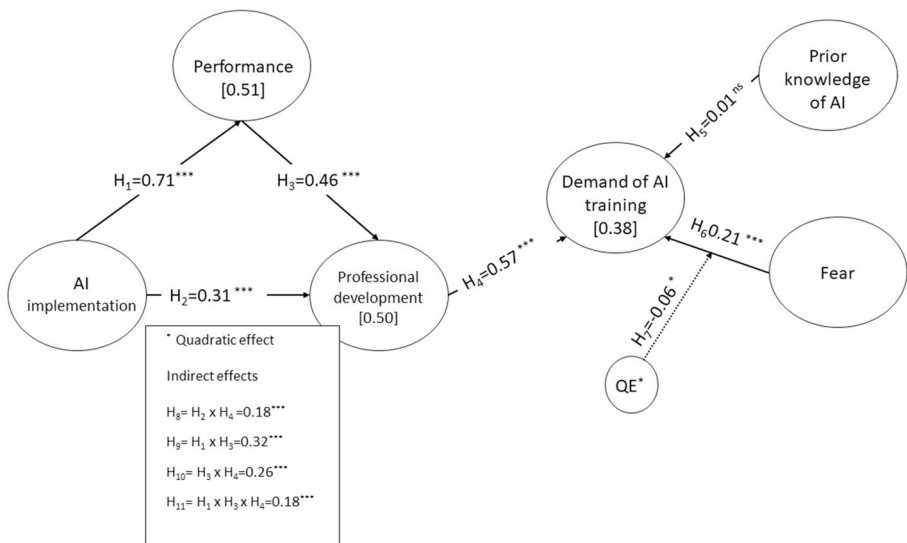


Fig. 2 Results. Source: Own elaboration

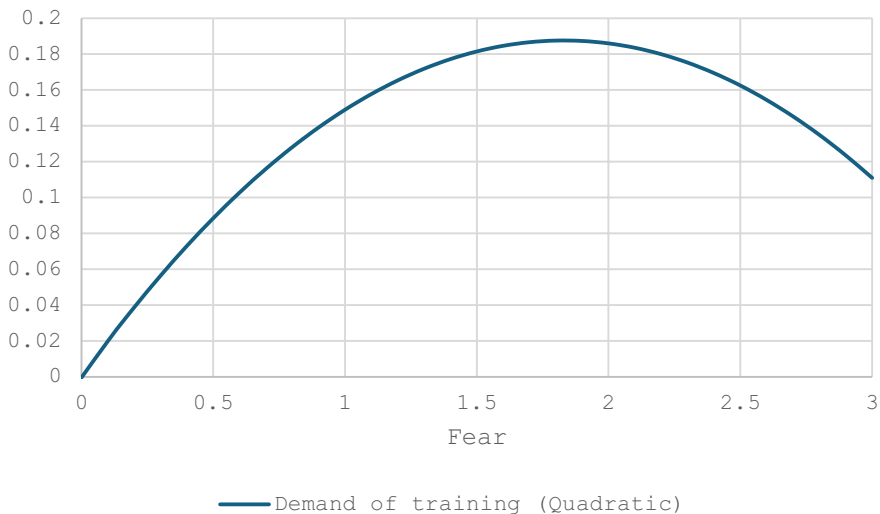


Fig. 3 Quadratic effect on the relationship between fear and demand of AI training. Source: Own elaboration

considered suitable. Similarly, the results demonstrated that the interaction term had a significant and negative direct effect on demand of AI training. An increase in fear by one standard deviation reduced the relationship with demand of AI training by 0.056. Thus, it can be concluded that the relationship between fear and demand of AI training diminishes exponentially at higher levels of fear, as can be seen in the Fig. 3. Indeed, Fig. 3 illustrates the diminishing marginal impact of fear on training demand, emphasizing the nonlinearity in emotional drivers of learning. Therefore, the hypothesis 7 is accepted.

6 Discussion

This study focused on students' perceptions of the impact of AI on business performance and professional development, addressing the shortcomings identified in previous research. The relevance and significant findings of the present work are evident through comparison with previous studies. The results of this study provide important insights into how students perceive artificial intelligence in the workplace and how those perceptions influence their academic motivations. From the perspective of Expectancy Value Theory, students appear to assign significant value to acquiring AI-related competencies, particularly when they believe these skills will enhance their employability and professional growth. The data suggest that the perceived performance of AI-driven companies positively shapes students' professional aspirations, thereby increasing their intention to pursue further academic training in AI.

However, this motivation is not uniform across the sample. The findings reveal a dual role of fear, consistent with literature on technology-related anxiety. On one hand, moderate levels of fear, such as concern about being replaced by machines, can

act as a motivational driver, prompting students to seek new skills. On the other hand, high levels of fear or self-doubt regarding their ability to understand AI may lead to avoidance behaviors or reluctance to engage with AI focused learning. This supports the conceptualization of fear as a multidimensional construct, where the emotional response to technology is not inherently negative, but its intensity and interpretation can significantly alter students' educational engagement.

These dynamics highlight the need for educational strategies that balance relevance (making AI feel professionally useful) with accessibility (reducing perceived difficulty and fear). By aligning the curriculum with students' expectations and addressing their concerns, academic institutions can better prepare students for AI-integrated workplaces.

Previous research has noted the importance of AI in improving business processes, data analysis and decision-making (Thaduri, 2020), however, there has been a dearth of studies exploring how students perceive these impacts (Nazaretsky et al., 2022). This study has bridged that gap by focusing on student perceptions, in comparison to studies such as Chan and Hu (2023) and Idroes et al. (2023), which also emphasise the influence of technology perceptions on the acceptance and use of new technologies.

Analyzing students' perceptions of AI reveals that most believe its implementation in companies significantly improves organizational performance. This result aligns with existing literature highlighting AI adoption as a factor of increased business efficiency and productivity (Abousaber & Abdalla, 2023; Glauner, 2020). Furthermore, students' perception of the positive impact of AI on their professional development is in line with previous studies highlighting the importance of high business performance on the quality of learning opportunities and professional development (Gault et al., 2010; Hergert, 2009).

The perception that entrepreneurial performance supports students' professional development aligns with theories such as Social Cognitive and Social Learning (Manz & Sims, 1981; Schunk, 2020) and Expectancy-Value and Experiential Learning (May, 2017; Montalbo, 2022). These results underline that high-performance environments improve students' practical skills and influence their psychological and motivational development (D'Abate, 2010; Gallardo et al., 2009).

As for prior knowledge, its impact on academic success in specific courses has been investigated (Hailikari et al., 2008; Hailikari & Nevgi, 2010; Seery, 2009), but how it influences the perceived need for additional AI training has not been sufficiently explored (Laupichler et al., 2022). Our results indicate that prior AI knowledge does not decrease the perceived need for additional training, which contrasts with previous studies suggesting that advanced knowledge in a specific area reduces the perceived value of additional academic training (Beatson et al., 2020; Wood & Lynch, 2002). One possible explanation for this discrepancy is the rapid evolution of AI technologies, which may lead students to perceive continuous learning as essential, regardless of their prior knowledge. Unlike other disciplines where foundational knowledge may be sufficient to sustain long-term expertise, AI requires ongoing updates due to its dynamic nature (Jha et al., 2022). Additionally, prior studies focused on fields where self-directed learning might be more effective, whereas AI education often requires structured learning environments to master complex algorithms and applications (Amershi et al., 2005; Aunimo & Huttunen, 2020). This aligns with constructiv-

ist theories, which suggest that while prior exposure facilitates learning, it does not necessarily reduce the perceived need for formal education when the subject matter is constantly evolving.

Furthermore, the relationship between fear of AI and demand for AI education has been studied in general terms (Alexandre et al., 2021; Hsu et al., 2023; Zucker & Gotian, 2022), but how extremely high levels of fear may inhibit interest in training has not been explored further (Beck & Clark, 1997; Fofana et al., 2020). Our findings suggest that even students with a strong background in AI perceive the need to continue their formal education to keep up with technological advances (Amershi et al., 2005; Jha et al., 2022). This study extends existing frameworks by integrating nonlinear emotional dynamics factors like fear into PLS-SEM models of technological implementation, a rarely explored area in AI adoption research.

The nonlinear effect of fear on AI training demand highlights a crucial psychological dynamic: while moderate levels of fear may serve as a motivational driver, excessive fear can have a paralyzing effect. This aligns with the anxiety processing theory (Beck & Clark, 1997), which suggests that heightened fear impairs rational decision-making, potentially discouraging students from engaging in AI education. Similarly, the inverted U-shaped relationship observed in our findings is reminiscent of the Yerkes-Dodson law (Yerkes & Dodson, 1908), which posits that an optimal level of stress enhances performance, whereas excessive stress leads to avoidance behaviors.

Finally, results showing that AI implementation in firms indirectly increases the demand for academic training through improved business performance and students professional development reflect existing literature indicating that AI adoption improves operational efficiency and, consequently, student's perception of the need to acquire advanced technological skills (Kishan Kumar, 2024; Zyryanova et al., 2024). Curricula should prioritize experiential AI learning to mitigate fear-induced mental blocks and capitalize on motivational drivers identified in this study.

The results of this study provide actionable insights for redesigning higher education curricula in response to students' perceptions of artificial intelligence in the workplace. Given that students link AI implementation in companies with both enhanced business performance and stronger career prospects, educational institutions must adapt their pedagogical approaches and curricular content accordingly.

First, AI literacy should be embedded as a transversal competency across university programs—not only in computer science or engineering, but also in business, social sciences, and humanities. This includes foundational knowledge about AI concepts, ethical implications, and real-world applications in different industries. As the study shows, students increasingly associate AI competence with employability, which should prompt universities to rethink program learning outcomes and include AI-related modules even in non-technical disciplines.

Second, teaching strategies should shift toward experiential learning approaches that simulate real AI-enabled business contexts. Project-based learning, interdisciplinary collaboration, and case study analysis can help students apply theoretical knowledge to practical challenges. This aligns with students' perceived need to prepare for AI-driven environments and helps reduce anxiety by fostering familiarity and confidence.

Third, faculty development and teacher training must be prioritized. For educational innovation to succeed, instructors themselves must become proficient in AI tools and pedagogical techniques that leverage AI in instruction and assessment. Institutions should provide structured support for educators to integrate AI meaningfully into their courses.

Finally, emotional aspects such as fear of AI should be explicitly addressed in educational settings. As the findings demonstrate, moderate fear may stimulate interest, but excessive anxiety becomes a barrier. Curriculum designers must integrate reflective activities and open discussions that allow students to explore concerns and build psychological readiness for AI adoption in professional contexts.

By aligning academic programs with students' expectations and technological realities, higher education can play a proactive role in shaping AI-aware, resilient, and ethically responsible future professionals.

7 Theoretical and practical implications

The results enrich the existing literature by offering a detailed view of how the implementation of AI in companies improves business performance, fosters students' professional development, increases their demand for academic training, and reduces fear of technology. By modeling complex mediating and moderating effects, this research elucidates how AI perceptions shape student trajectories, offering a blueprint for aligning educational strategies with technological advancements.

This study offers critical theoretical contributions by exploring students' perceptions of the impact of artificial intelligence on business and their professional and professional development. The proposed model provides a comprehensive understanding of how implementing artificial intelligence is perceived as a catalyst for students' business performance and professional development. Furthermore, it establishes the indirect connection between good business performance and the demand for academic training, mediated by students' professional development. These findings enrich the literature on adopting emerging technologies and their influence on human capital, providing a solid theoretical basis for future research.

From a practical perspective, the implications of this study are highly relevant for business managers and education policymakers. Students' perception that the implementation of AI in companies improves corporate performance and contributes to their professional development suggests the need to encourage the adoption of advanced technologies in the business environment. Managers should consider AI as a strategic tool that optimises operations and enriches human capital, enhancing their employees' professional growth.

The results of this study offer valuable implications for practitioners, policymakers, and stakeholders in AI and education. Education practitioners can benefit from integrating AI into curricula, as students' perceptions suggest that exposure to AI enhances business performance and professional development, better preparing graduates for today's labour market. For business managers, the findings highlight AI as a critical driver of organizational performance, improving efficiency and competitiveness while positively impacting employees' professional growth. This fos-

ters a culture of continuous learning essential in a dynamic business environment. Policymakers are encouraged to support AI adoption in businesses and educational curricula through tax incentives and initiatives that promote collaboration between the private sector and educational institutions, enhancing economic competitiveness and workforce readiness.

This study highlights the crucial role of how AI-related risks are communicated to students. The nonlinear relationship between fear and AI training demand suggests that while moderate concern can encourage learning, excessive fear may lead to avoidance. Overemphasizing automation-driven job displacement, for instance, could create counterproductive anxiety, deterring students from acquiring AI-related skills. To address this, AI education programs should not only focus on technical knowledge but also integrate reassuring narratives that emphasize adaptability, skill acquisition, and human-AI collaboration.

Educational institutions and policymakers must design learning environments that reduce emotional barriers to AI adoption. By fostering a balanced understanding of AI—acknowledging both its challenges and opportunities—students can develop confidence in their ability to navigate an AI-driven job market. This will not only enhance their participation in AI training but also contribute to a broader and more effective adoption of artificial intelligence in society.

8 Conclusions

This article contributes significantly to understanding students' perceptions regarding implementing AI in companies and its effects on organisational performance and professional and academic development. The research confirms that students perceive the adoption of AI to have a positive impact not only on company performance but also on their own professional and academic development.

The importance of this analysis lies in its ability to elucidate how AI, viewed through the lens of future professionals, can be a driving force for change in education and the labour market. This student perspective is crucial, as these individuals are the future employees and leaders who will guide the integration of new technologies in organisations.

Based on the study's findings, several recommendations can be made for educational policymakers and industry leaders. For policymakers, integrating AI into curricula, promoting digital literacy, addressing AI-related anxieties, and fostering collaboration between academia and industry are crucial. Industry leaders should embrace AI as a strategic tool, prioritize employee training, adhere to ethical guidelines, and collaborate with educational institutions. By implementing these recommendations, both groups can contribute to a future where AI is used ethically and effectively, benefiting society and individuals alike.

From a teaching innovation perspective, the findings indicate that academic curricula should be redesigned to systematically integrate AI literacy into subject planning, incorporating practical modules and project-based assessments that simulate real AI business environments. In addition, it is essential to implement ongoing teacher training plans that include digital skills development and active pedagogical

methodologies, such as experiential learning and formative assessment, to ensure that teachers are prepared to facilitate the ethical and practical use of AI in the classroom.

A study analyzing university students' perceptions of the impact of AI in the business sector on their educational and professional development, using a sample from specific disciplines at three universities during a single point in time, faces several limitations. The sample is limited to students, which may not represent other stakeholders' perceptions, such as working professionals or educators. The findings may lack generalizability due to geographic, disciplinary, and temporal constraints, and may not reflect broader or evolving perspectives. Selection bias from the choice of universities and student profiles could further skew results, as could limited diversity in demographic factors. Self-reported data risks inaccuracies, and external variables, such as regional economic conditions and curriculum design, may not be adequately accounted for. Finally, the study's cross-sectional nature and potential influence of current trends could lead to a narrow understanding of a complex and dynamic issue. Exploring other variables, such as organisational culture or readiness for change, would provide a deeper understanding of AI acceptance. Therefore, future research should investigate cross-cultural perceptions of AI using panel data to validate these findings, and longitudinally track how student perceptions of AI evolve as they transition into professional roles, examining the interplay of workplace readiness and educational interventions.

From an educational perspective, the study emphasises the pressing requirement to redesign higher education curricula in line with students' expectations concerning artificial intelligence. Academic programmes must incorporate AI literacy as a core competence across disciplines, with modules covering both technical knowledge and ethical reflection. Pedagogical approaches should shift towards experiential and interdisciplinary learning, enabling students to engage with real-world AI applications. Furthermore, the emotional dimensions revealed in this study, particularly fear, call for curricular strategies that foster psychological readiness for AI integration, as well as transferring knowledge. Ultimately, the findings support the development of inclusive, adaptive and forward-looking educational models that will prepare students to thrive in AI-driven professional environments.

Appendix: perception of artificial intelligence

Artificial Intelligence perception

Indicate from 1 to 5, where 1 is not at all and 5 is very much, the degree of implementation you believe AI will have in the following business departments.

	1	2	3	4	5
Human Resources	☐	☐	☐	☐	☐
Finance	☐	☐	☐	☐	☐
Management	☐	☐	☐	☐	☐

	1	2	3	4	5
Accounting	☐	☐	☐	☐	☐
Production	☐	☐	☐	☐	☐
Sales	☐	☐	☐	☐	☐
Supply	☐	☐	☐	☐	☐
Marketing	☐	☐	☐	☐	☐
Research and Development	☐	☐	☐	☐	☐

Benefits of Artificial Intelligence in Business

Indicate from 1 to 5, where 1 means strongly disagree and 5 means strongly agree, your level of agreement with the following statements:

	1	2	3	4	5
AI will increase the sales of companies	☐	☐	☐	☐	☐
AI will increase the profitability of companies	☐	☐	☐	☐	☐
AI will increase the return on capital employed	☐	☐	☐	☐	☐
AI will improve the quality of products and services provided	☐	☐	☐	☐	☐
AI will enable companies to develop new products and services	☐	☐	☐	☐	☐

Personal Perception

Indicate from 1 to 5, where 1 means strongly disagree and 5 means strongly agree, your level of agreement with the following statements:

	1	2	3	4	5
I believe that my professional future will be influenced by AI	☐	☐	☐	☐	☐
I should train more in AI	☐	☐	☐	☐	☐
I have sufficient knowledge about AI	☐	☐	☐	☐	☐
The development of AI makes me fearful	☐	☐	☐	☐	☐
The implementation of AI represents a professional stimulus for me	☐	☐	☐	☐	☐
AI will provide a professional advantage for me	☐	☐	☐	☐	☐

Funding Open Access funding provided thanks to the CRUE-CSIC agreement with Springer Nature.

Data availability The authors declare that the data on which the article is based will be made available to any reviewer or editor on request.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests

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